

Trainings ▾

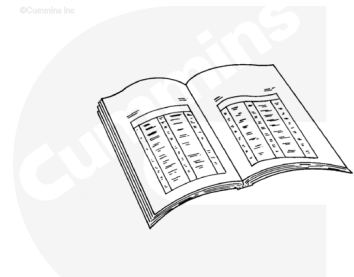
Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



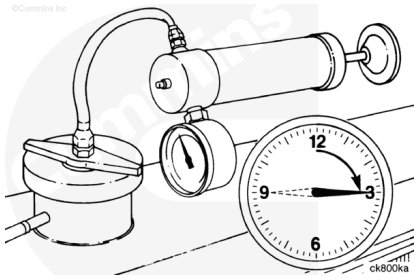
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- Drain the engine lubricating oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/10/10-007-037-tr.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/10/10-007-025-tr.html)
- Remove the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7.
(/qs3/pubsys2/xml/en/procedures/10/10-007-035-tr.html)
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/10/10-001-089-tr.html)

Leak Test

Pressurize the engine cooling system. Refer to Procedure 008-018 in Section 8 (/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html).

Note : Apply the air pressure 15 minutes before inspecting the cylinder liner seats. Make sure the system is holding air pressure before beginning inspection.



Measurements

	kpa	psi
Air Pressure:	138	20

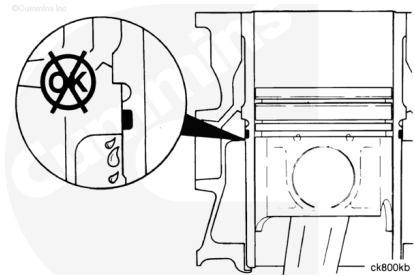
Inspect the outside circumference of the cylinder liners and the area below the cylinder liner seats in the cylinder block for coolant leaks.

If a leak is found:

- Remove and inspect the cylinder liners. Refer to Procedure 001-028 in Section 1. (/qs3/pubsys2/xml/en/procedures/10/10-001-028-tr.html)

Refer to the Cylinder Liner Counterbore Ledge Reuse Guidelines, Bulletin 4383753, (/qs3/pubsys2/xml/en/bulletin/4383753.html) to inspect the counterbore ledge surface areas for signs of pitting, fretting, or wear.

- Inspect the cylinder block liner bore area. Refer to Procedure 001-058 in Section 1. (/qs3/pubsys2/xml/en/procedures/10/10-001-058-tr.html)



Finishing Steps

- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1. (/qs3/pubsys2/xml/en/procedures/10/10-001-089-tr.html)
- Install the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7.



(/qs3/pubsys2/xml/en/procedures/10/10-007-035-tr.html)

- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/10/10-007-025-tr.html)
- Fill and prime the lubricating oil system. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/10/10-007-037-tr.html)
- Operate engine to normal operating temperature and check for leaks.

If a malfunction resulted in coolant, oil, excessive fuel, or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected. Refer to Procedure 014-013 in Section 14.

(/qs3/pubsys2/xml/en/procedures/101/101-014-013-tr.html)